

$\gamma\delta$ T cells and epigenetic drugs: A useful merger in cancer immunotherapy?

Jaydeep Bhat and Dieter Kabelitz*

Institute of Immunology; University of Kiel and UKSH Campus Kiel; Kiel, Germany

$\gamma\delta$ T cell-based immunotherapeutic strategies in cancer patients are as yet of limited success. Drugs targeting epigenetic mechanisms including histone acetylation and DNA methylation trigger cell death in tumor cells but in addition have immunomodulatory activity. Here, we discuss the potential benefit of combining both strategies in cancer immunotherapy.

$\gamma\delta$ T Cells: Limited T-Cell Receptor Diversity, Multiple Tumor Target Molecules

In view of their unique features, $\gamma\delta$ T cells have recently raised substantial interest as a cellular target in cancer immunotherapy. Major subsets of human $\gamma\delta$ T cells can be categorized on the basis of their expressed V γ /V δ gene usage. V γ 9V δ 2 T cells (hereafter termed V δ 2 T cells) dominate in the peripheral blood where they usually account for 50 to 95% of all $\gamma\delta$ T cells.¹ $\gamma\delta$ T cells expressing other V δ genes (mostly V δ 1 or V δ 3) that can be paired with various V γ elements are here collectively termed non-V δ 2 T cells. These non-V δ 2 T cells are usually localized in the mucosal tissues where they constitute a substantial proportion of intraepithelial lymphocytes.² $\gamma\delta$ T cells kill a broad range of tumor cells of epithelial origin as well as many leukemia/lymphoma cells in a T-cell receptor and/or Natural Killer Group 2 Member D (NKG2D)-dependent manner.³⁻⁶ V δ 2 T cells recognize tumor cells due to their enhanced production of isopentenyl pyrophosphate (IPP), an intermediate of the mevalonate pathway which is frequently dysregulated in tumor cells.⁷ Importantly,

the IPP production can be easily manipulated by nitrogen-containing bisphosphonates (N-BP), drugs that are widely used in the clinics for the treatment of bone fragility disorders. N-BP inhibit an enzyme of the mevalonate pathway downstream of IPP synthesis, thereby inducing accumulation of IPP and enhanced sensitivity of tumor cells to V δ 2 T cell cytotoxicity.⁸ In addition to the T-cell receptor-dependent recognition of IPP, tumor cell killing by V δ 2 T cells can be triggered through antibody-dependent cellular cytotoxicity⁹ and tumor-targeting bispecific antibodies.¹⁰ Tumor surface-expressed antigens that are specifically recognized by the T-cell receptor of non-V δ 2 T cells include the endothelial protein C receptor¹¹ and constitutively expressed NKG2D ligand MICA.¹² Both V δ 2 and non-V δ 2 T cells are thus of interest for cancer immunotherapy.¹³

Clinical Application of Human $\gamma\delta$ T Cells

Two strategies to exploit $\gamma\delta$ T cells for immunotherapy can be envisaged, i.e., the *in vivo* activation of $\gamma\delta$ T cells by selective stimulation, or the adoptive transfer of *in vitro* expanded $\gamma\delta$ T cells. Wilhelm *et al.* were the first time to attempt *in vivo* activation of $\gamma\delta$ T cells by giving N-BP (pamidronate) plus low-dose IL-2 to patients with lymphoid malignancies including relapsed/refractory low-grade non-Hodgkin lymphoma or multiple myeloma. They demonstrated the feasibility of this approach and reported some anti-lymphoma efficacy.¹⁴ Five years later, Kondo *et al.*¹⁵ established an efficient protocol for large-scale *ex vivo* expansion of

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Abbreviations: DNMTi, DNA methyltransferase inhibitor(s); HDACi, histone deacetylase inhibitor(s); IL-2, interleukin-2; N-BP, nitrogen-containing bisphosphonate; NKG2D, natural killer group 2 member D; NSCLC, non-small cell lung cancer

*Correspondence to: Dieter Kabelitz; Email: dietrich.kabelitz@uksh.de

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$\gamma\delta$ T cells from patients with advanced non-small cell lung cancer (NSCLC), bone metastatic breast or prostate cancer, or lung metastatic colorectal cancer. These two lines of clinical application have been explored by several groups since then in clinical phase I/II studies, including the combination of *in vivo* activation plus adoptive transfer of $\gamma\delta$ T cells (Fig. 1A).

In vivo activation of $\gamma\delta$ T cells by N-BP (mostly zoledronate) plus low-dose IL-2 has been investigated in various cancer entities. In a phase I clinical trial with zoledronate and IL-2 in patients with metastatic hormone-refractory prostate cancer, Dieli *et al.*¹⁶ observed *in vivo* activation of V δ 2 T cells that produced TNF-Related Apoptosis Ligand (TRAIL). In accordance, increased serum levels of

TRAIL were present in the serum, and clinical responses were observed in some of the patients. Using a similar zoledronate plus low-dose IL-2-based protocol in a phase I study in advanced metastatic breast cancer, Meraviglia *et al.* observed a strong correlation between a clinical response (i.e., reduction of carcinoma progression) and the peripheral V δ 2 T cell numbers.¹⁷ A similar approach was extended to patients with metastatic renal cell carcinoma. Interestingly, however, the repeated administration of zoledronate and IL-2 reduced the proliferative capacity of V δ 2 T cells.¹⁸ This might in fact be related to an inhibitory effect of granulocytes on V δ 2 T cell activation that we observed *in vitro* upon uptake of N-BP by granulocytes.¹⁹ Taken together,

administration of zoledronate plus low-dose IL-2 emerges as a novel, safe and feasible approach for $\gamma\delta$ T cell-based immunotherapy. However, treatment regimens need to be optimized for the therapeutic *in vivo* augmentation of V δ 2 T cells. Moreover, potential adverse effects of the *in vivo* activation of V δ 2 T cells need to be carefully considered.²⁰

The adoptive transfer of zoledronate plus (in this case high dose) IL-2-activated and *in vitro* expanded $\gamma\delta$ T cells was first applied by Abe *et al.*²¹ to six patients with multiple myeloma. No serious treatment-related adverse effects were observed. A detailed analysis revealed an increase in CD27⁺CD45RA⁺ effector memory (TEM) V δ 2 T cell subset in the patients, and soluble NKG2D ligand MICA was

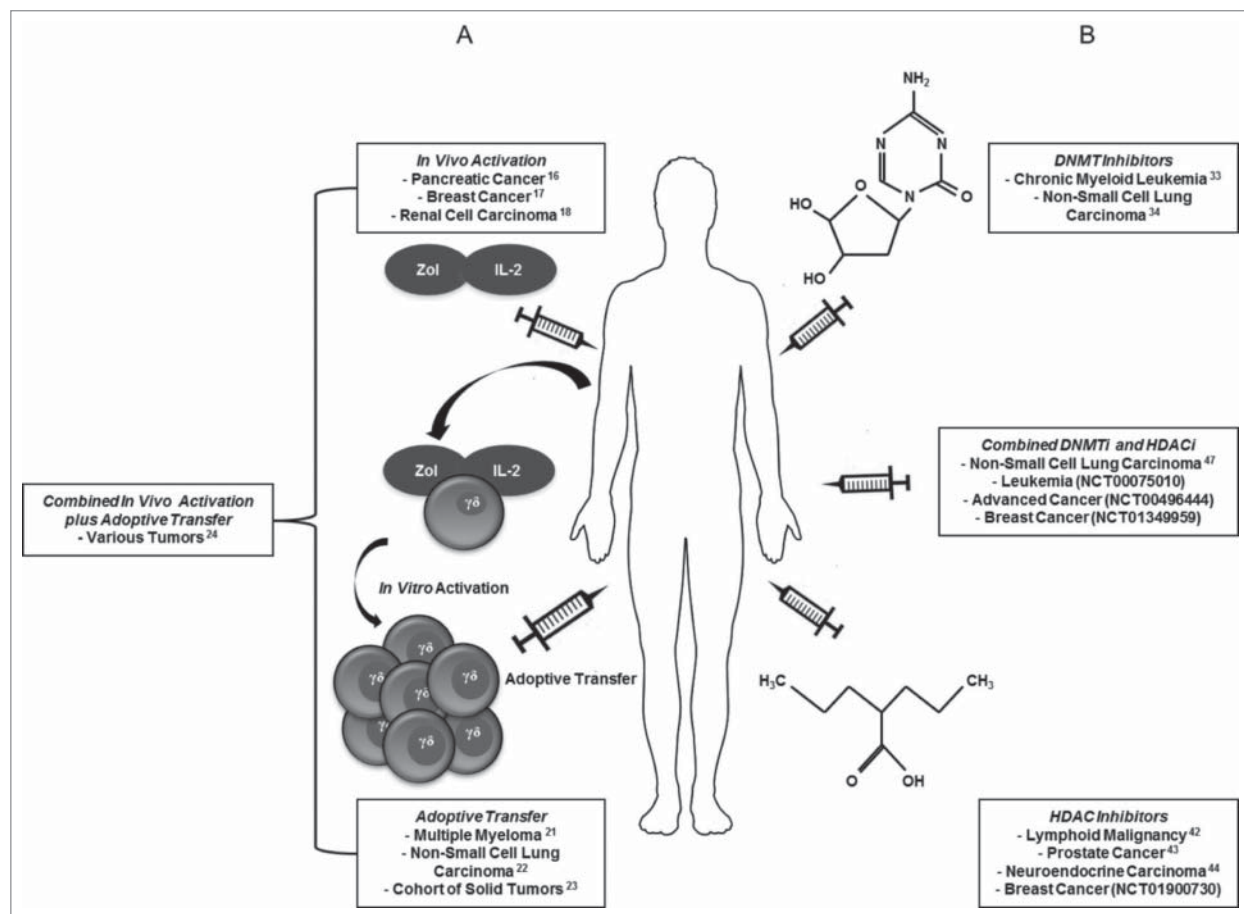


Figure 1. An overview of human $\gamma\delta$ T cell-based immunotherapies and epigenetic inhibitors used in the clinics. **(A).** Clinical studies in patients with various malignancies are based on *in vivo* activation of $\gamma\delta$ T cells (top) or adoptive immune transfer (bottom), or a combination thereof (left). **(B).** Use of DNMT (top) and HDAC inhibitors (bottom) for monotherapy or as a combination therapy (middle) in cancer patients. Ongoing or finished clinical trials are indicated with references, otherwise with respective identifiers adopted from US National Institutes of Health website (<http://ClinicalTrials.gov>) as of December 12, 2014. Zol, zoledronate.

detected in the serum of some patients with elevated myeloma (M-) protein. Other studies performed e.g., in patients with NSCLC also supported the safety and feasibility of adoptive transfer of $\gamma\delta$ T cells.²² Importantly, another phase I study in patients with advanced solid tumors performed by Nicol *et al.*²³ provided preliminary clinical evidence for the migration of adoptively transferred $\gamma\delta$ T cells to tumor sites. A systematic review of all clinical trials with $\gamma\delta$ T cells using both treatment options (i.e., *in vivo* activation and/or adoptive transfer) was performed by Fisher *et al.*²⁴ Their analysis of pooled data of 12 clinical trials with 157 patients highlighted the promising clinical efficacy of human $\gamma\delta$ T cell immunotherapy. It is obvious, however, that a therapeutic benefit of $\gamma\delta$ T cell-targeted immunotherapy might best be expected in combination with other therapeutic strategies.²⁵

Shedding of NKG2D ligands from the tumor cell surface is a known tumor escape mechanism, due to the inhibitory effect of soluble NKG2D ligands on NKG2D receptor expressing killer cells including $\gamma\delta$ T cells.^{26,27} There is substantial heterogeneity in the mechanisms and the extent of NKG2D ligand shedding among various tumor entities, thus making a prediction of the consequences with respect to $\gamma\delta$ T cell therapy in a given patient difficult.²⁸ Therefore, a personalized approach with novel treatment regimens might be required to enhance the efficacy of $\gamma\delta$ T cell-based therapy.

Epigenetic Modifiers in The Clinic

The epigenetic modification of protein expression based on the inhibition of enzymes that introduce and remove chromatin modifications is an established treatment option in oncology. Enzymes involved in chromatin modifications include DNA methyltransferases (DNMTs), histone lysine acetyltransferases (KATs), histone lysine methyltransferases (KMTs), histone lysine demethylases (KDMs), histone deacetylases (HDACs), protein arginine methyltransferases (PRMTs) as well as proline isomerases, kinases and ubiquitylases,

many of which are represented by families of proteins.²⁹

One of the well-established epigenetic inhibitors used in clinical and basic research is 5-azacytidine (Aza-CR) and its derivative 5-aza-2'-deoxycytidine (5-AzaCdR). Both were synthesized by Piskala and Sorm³⁰ and were tested as nucleoside antimetabolites with clinical application in acute myelogenous leukemia.³¹ Later on, Flatau *et al.*³² demonstrated that the concentration-dependent cytotoxic effect is related to the effects on DNA methylation. 5-AzaCdR (decitabine) is an established drug for the therapy of leukemias. In a study involving 130 patients with chronic myelogenous leukemia (CML), the therapeutic efficacy and toxicity of decitabine were analyzed in different stages of CML.³³ Therapy with decitabine has also been extended to certain solid tumors. Several studies in patients with NSCLC have shown remarkable therapeutic efficacy.^{34,35} Based on available preclinical and clinical data, Karahoca and Momparler³⁶ proposed a novel dose regimen for the use of decitabine in oncology. The molecular mechanism responsible for the efficacy varies substantially from inhibitor to inhibitor.³⁷ Among others mechanism, decitabine induces apoptosis but also promotes differentiation, senescence and autophagy.³⁸ Hence, after extensive studies on efficacy and safety, various analogs of DNMTs inhibitors have been recently approved as antitumor agents.

Most commonly known as antiepileptic drug, mood stabilizing agent and anti-convulsant drug, the prototypic HDAC inhibitor valproic acid represents another class of therapeutic epigenetic drugs. The generic name of valproic acid is Dipropylacetate (DPA), synthesized in 1881. However, its biological effect on leukocytes and platelets was studied in Sprague-Dawley Rats only in 1985.³⁹ Several studies have shown that class I HDAC inhibitors such as valproic acid increase NK cell mediated killing of tumor cells primarily *via* the NKG2D-NKG2D ligand axis.^{40,41} Valproic acid exerts antitumor activity on multiple lymphoid malignancies like B cell-, T cell- and NK cell- lymphomas. The study revealed increased efficacy when combined with

the proteasome inhibitor Bortezomib.⁴² Based on these results, valproic acid has been studied many fold in clinical trials. Ten patients with Castration-Resistant Prostate Cancer (CRPC) were included in a phase II trial performed by Sharma *et al.*⁴³ This study revealed intolerance of valproic acid administration with significant levels of toxicity. The authors concluded that oral application of valproic acid should be avoided in CRPC patients. In contrast, however, valproic acid resulted in a tumor response and increased survival associated with enhanced tumor marker production and Notch1 activation in a pilot phase II study of low-grade neuroendocrine carcinoma.⁴⁴ In view of such discrepancy, an interventional, randomized phase II study is currently ongoing in patients with breast cancer (NCT01900730). In this study, valproic acid is expected to stop cancer cell division and differentiation, thereby ultimately leading to less pleural fluid production. Considering the diverse molecular and physiological effects of epigenetic modifiers, treatment protocols have been designed combining potent inhibitors of both DNMT and HDAC (Fig. 1B).

Along this line, several reports revealed superior effects of combined epigenetic inhibitor application as opposed to single agents. In a phase I/II study in patients with advanced leukemia, the combination of 5-AzaCdR with valproic acid was safe and effective, and was associated with a transient reversal of aberrant epigenetic markers (NCT00075010).⁴⁵ Leclercq *et al.*⁴⁶ studied the anticancer effect of 5-AzaCdR combined with valproic acid in an aggressive tumor, malignant pleural mesothelioma. In addition to tumor cell death, they observed immunomodulatory effects at the level of cytokine production including interferon- γ and TGF β expression of CD8⁺ cytotoxic T lymphocytes. A non-randomized, phase I study of low-dose DNMT inhibitor 5-Azacytidine (Aza-CR) combined with HDAC inhibitor valproic acid in patients with advanced cancers was recently performed to evaluate adverse effects and maximum tolerated doses (Razelle Kurzrock, MD Anderson Cancer Centre, NCT00496444). The combination of decitabine and valproic acid turned out to be effective in patients

with NSCLC, with limited neurological toxicity at relatively high dosage.⁴⁷ Furthermore, a phase II study of Aza-CR in combination with HDAC inhibitor entinostat is ongoing in women with advanced triple negative or hormone-refractory breast cancer (Vered Stearns, Sidney Kimmel Comprehensive Cancer Centre, Australia and National Cancer Institute, USA; NCT01349959).

Taken together, targeting different pathways of epigenetic modification (DNMT, HDAC) promises to be more effective in cancer therapy than the inhibition of single pathways.

$\gamma\delta$ T Cells and Epigenetic Modifiers – A Useful Liaison?

Cellular processes including differentiation and cell type-specific functions are determined by the digital language of covalent histone modification. A multitude of modifications act as guide for every cell fate decision, for sensing of microenvironmental signals and for cellular adaptation under pathophysiological conditions.⁴⁸ Not surprisingly, this also applies to T cell differentiation. Accordingly, histone modification marks characterize early T cell development from multipotent progenitors to committed T cells,⁴⁹ primary human leukocyte differentiation into CD4⁺, CD8⁺, CD14, CD16, CD19 subsets,⁵⁰ memory T cell differentiation,^{51,52} but also transition from normal lymphocyte development into neoplastic transformation and lymphomagenesis.⁵³

With regard to tumor immunotherapy, the potential application of epigenetic drugs needs to be carefully evaluated. For instance, 5-Aza-2'-deoxycytidine or decitabine efficiently demethylate FoxP3 CpG loci and thereby induce a strong and stable suppressive activity of Treg which is unwanted in the context of cellular antitumor immunity.⁵⁴ On the other hand, class I HDAC inhibitor entinostat was found to reduce FoxP3 expression and to reduce suppressive Treg activity associated with better efficacy of immunotherapies in murine cancer models.⁵⁵ Moreover, HDAC inhibitors modulate CD4⁺ T cell polarization *in vitro* and *in vivo*⁵⁶ which

again might have an impact on tumor immunity. So far, limited information is available on the epigenetic signature of $\gamma\delta$ T cells. A recent study has shed some light on the epigenomic analysis of IL-17-producing CD27[−] and IFN γ -producing CD27⁺ murine $\gamma\delta$ T cells. The detailed analysis emphasizes the effect of subset-specificity on key transcriptional regulators of differentiation and peripheral function in murine thymus-derived $\gamma\delta$ T cells.⁵⁷ It is also known that IFN γ production in splenic $\gamma\delta$ T cells is controlled through hypomethylation with simultaneous requirement for Eomes and T-bet transcription factors.⁵⁸ Our own preliminary studies reveal an overall differential methylation status of human peripheral blood $\gamma\delta$ T cells as compared to $\alpha\beta$ T cells. Furthermore, we noted a selective modulation of some surface markers when activated human $\gamma\delta$ T cells are treated with valproic acid (Bhat *et al.*, unpublished).

Given the constitutive expression of the activating NKG2D receptor on virtually all $\gamma\delta$ T cells, enhanced and/or stable expression of NKG2D ligands on tumor cells should greatly facilitate their killing by $\gamma\delta$ T cells. NKG2D ligands are highly methylated and thus transcriptionally suppressed in many lymphoma and tumor cells.⁵⁹ Consequently, treatment with 5-aza or decitabine restores NKG2D ligand expression.⁵⁹ Similarly, NKG2D ligand expression is induced or enhanced on diverse tumor entities including liver cancer, osteosarcoma, pancreatic carcinoma, myeloma, leukemias by HDAC inhibitors.^{60–64} Importantly, increased surface expression of NKG2D ligands occur in the absence of enhanced shedding which might negatively interfere with NKG2D receptor activation.⁶² Thus, we consider the increase of NKG2D ligand expression by epigenetic drugs a straightforward and doable approach to enhance $\gamma\delta$ T cell immunotherapy.

At this stage it is unclear, however, how epigenetic drugs will affect the functional program of human $\gamma\delta$ T cells, an issue that requires further investigation. NKG2D is demethylated in NK cells and CD8⁺ T cells (and presumably in $\gamma\delta$ T cells as well, although this was not explicitly studied here).⁶⁵ Importantly,

treatment with the histone acetyltransferase inhibitor Curcumin reduced NKG2D transcription and reduced the lytic capacity of a NK cell line,⁶⁵ which again would be a serious disadvantage in the setting of $\gamma\delta$ T cell immunotherapy. Based on all these considerations, we believe that treatment of cancer patients with epigenetic drugs combined with an adoptive transfer of *in vitro* expanded $\gamma\delta$ T cells currently holds the best promise for a useful liaison.

Concluding Remarks

$\gamma\delta$ T cells are important players in cellular immunotherapy. Accumulating evidence suggests that $\gamma\delta$ T cell immunotherapy needs to be combined with other strategies for improved clinical results. Among others, these additional strategies might include chemotherapeutics, antibody constructs, and epigenetic drugs. While the impact of epigenetic drugs on human $\gamma\delta$ T cell transcriptional regulation and functions needs to be investigated, the proposed two-step approach (treatment with epigenetic drugs plus adoptive transfer of *in vitro* expanded $\gamma\delta$ T cells) is ready for clinical evaluation.

Disclosure of Potential Conflicts of Interest

This work forms part of the PhD thesis of JB. There are no other potential conflicts of interest.

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